

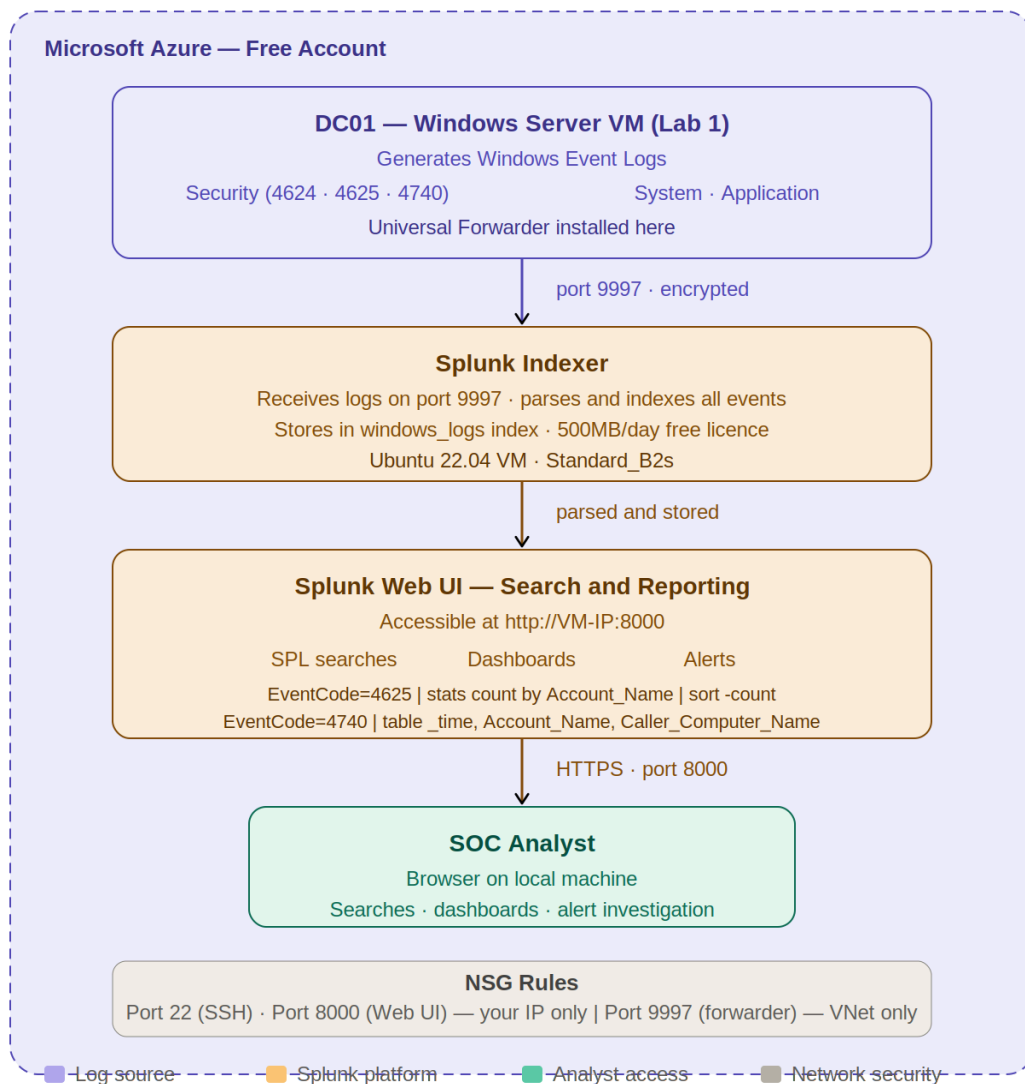
Lab 3 — Splunk SIEM & Log Analysis

Splunk Free · Azure VM · SOC Skills · Security Monitoring

Field	Value
Certification alignment	CompTIA Security+ · CySA+ · Splunk Core Certified User
Free tools	Splunk Enterprise — 60-day trial then permanently free with 500MB/day
Time to complete	4–6 hours across multiple sessions
Estimated cost	\$0 — Splunk Free licence covers everything in this lab
Career relevance	SOC Analyst (Tier 1–3) · Security Engineer · Incident Responder

Architecture — How Splunk receives and processes logs

The diagram below shows the complete data flow for this lab. Your Windows Server VM from Lab 1 forwards logs to a separate Ubuntu VM running Splunk. Splunk indexes those logs and makes them searchable through the web UI.



The Business Problem This Lab Solves

A medium-sized organisation generates millions of log events every day — Windows Event Logs from workstations, authentication logs from Active Directory, firewall logs from network equipment, web server access logs, cloud resource logs. Without a SIEM, those logs sit in separate systems and nobody can search across them, correlate events, or identify patterns that indicate an attack.

The SIEM is the security operations centre's primary tool. When an alert fires, the SOC analyst opens the SIEM and searches the logs to understand what happened, when, from where, and what was affected. When a security team wants to know whether a particular threat has touched their environment, they search the SIEM.

Splunk is the most widely deployed commercial SIEM. Learning it in a free lab environment gives you a concrete, demonstrable skill that appears on job descriptions for almost every security operations role.

Role	How this lab applies
SOC Analyst Tier 1	Monitoring dashboards for alerts, searching logs for suspicious activity, escalating findings
SOC Analyst Tier 2–3	Building detection rules, correlating events across data sources, threat hunting
Cloud Security Engineer	Microsoft Sentinel and AWS Security Hub use the same SIEM concepts — this lab teaches the mental model
Incident Responder	Searching logs during an active incident, building a timeline of events, identifying scope of compromise

Key Concepts — Read This Before Starting

What is a SIEM?

SIEM stands for Security Information and Event Management. A SIEM is a platform that collects log data from across your entire environment — servers, workstations, firewalls, cloud services, applications — and makes it all searchable in one place. The two core jobs of a SIEM are: (1) correlation — connecting events across different systems to identify patterns that no single system would reveal alone, and (2) alerting — automatically notifying analysts when suspicious conditions are met. Without a SIEM, a security team would need to log into each system individually and search its logs manually. With a SIEM, one analyst can search across thousands of systems simultaneously.

What is SPL — Splunk Processing Language?

SPL is the query language you use to ask Splunk questions. It works as a pipeline — you start with a search, then pipe the results through commands that filter, transform, and visualise the data. For example: `index=windows_logs EventCode=4625 | stats count by Account_Name | sort -count`. This search finds all failed login events, counts them grouped by username, and sorts highest to lowest. Every SPL search follows the same pattern: find the events, then shape the results. You do not need to know SPL before starting this lab. Each search in the steps below is explained line by line.

What is a Splunk index?

An index in Splunk is like a database table — it is a named storage bucket where events are kept. When logs arrive at Splunk, they are stored in an index. When you search, you specify which index to search with `index=name`. You can have multiple indexes for different data sources: one for Windows logs, one for firewall logs, one for web server logs. Keeping data in separate indexes lets you control retention, permissions, and storage separately for each data source. In this lab you will create one index called `windows_logs`.

What is the Universal Forwarder?

The Universal Forwarder is a lightweight agent that Splunk provides for free. You install it on any machine whose logs you want to send to Splunk. It monitors log files and Windows Event Logs, compresses and encrypts the data, and forwards it to your Splunk indexer over port 9997. The forwarder uses very little CPU and RAM — it is designed to run invisibly in the background on production servers. In this lab you install it on your Windows Server VM from Lab 1 so that Active Directory security events flow into your Splunk instance automatically.

What is Windows Event IDs?

Windows records everything that happens on a system as numbered events in the Windows Event Log. Each type of event has a unique ID number. The three IDs you will use constantly in security work are: EventCode 4624 — a user successfully logged on. EventCode 4625 — a logon attempt failed. EventCode 4740 — a user account was locked out. When a SOC analyst investigates a potential breach, these three event codes tell the story: who tried to log in, whether they succeeded, and whether accounts got locked out along the way. A spike in 4625 events for one account in a short window is a brute force attack. Hundreds of 4625 events spread across many accounts is a password spray attack.

What is inputs.conf?

inputs.conf is a configuration file that tells the Splunk Universal Forwarder what data to collect. It lives on the machine where the forwarder is installed — in this lab that is your Windows Server VM. Each section in the file, marked with square brackets, defines one data source. [WinEventLog://Security] tells the forwarder to collect Windows Security events. The settings underneath control how: disabled=0 means enabled, start_from=oldest means collect historical events not just new ones. You edit this file once during setup and the forwarder reads it automatically. You do not need to restart the forwarder every time you add a new log source — just edit the file and restart once.

What You Will Learn

Skill	Real-world application
Deploy Splunk and configure a data input	Every Splunk deployment starts with getting data in — the Universal Forwarder is how most enterprise environments feed logs to Splunk
Navigate the Splunk interface	Search, dashboards, alerts, reports — understanding the layout is table stakes for any SOC role
Write SPL searches	SPL is how you ask Splunk questions. It is the skill that separates analysts who find threats from analysts who stare at dashboards
Build security dashboards	Visualising security data at a glance — login failures over time, top source IPs, failed authentication by user

Identify failed login attempts	One of the most common security investigations. Distinguishing normal user error from a brute force attack
Build an automated alert	Splunk fires an alert when conditions you define are met, rather than waiting for a human to notice
Search for account lockout events	A trail of lockout events can indicate a password spray attack in progress

Step 1 — Get Splunk Free

Splunk Enterprise is free to download. You get a 60-day full trial, after which it automatically converts to the free licence. The free licence limits you to 500MB per day of data indexing — more than enough for a home lab.

When you go to the Splunk website you will see several products listed. You want Splunk Enterprise — not Splunk Cloud (that is a hosted SaaS product) and not Splunk SOAR. Look for the download labelled Splunk Enterprise and select the version for your operating system. To download, Splunk requires you to create a free account and fill in a registration form. You do not need to use your real email address or personal details for this — use a temporary email address and dummy information instead (see the note below).

1. Go to splunk.com/en_us/download/splunk-enterprise.html
2. Create a free Splunk account using a temporary email address — see the note below for exactly how to do this
3. Download Splunk Enterprise for Linux (the Ubuntu VM you will create below)
4. Follow the VM setup and install instructions in the next section

Use a temporary email address for your Splunk account

Splunk requires you to fill in a registration form before you can download. You do not need to use your real personal details. Use a free temporary email service and fill in the rest of the form with dummy information.

Step 1 — Get a temporary email address:

Go to temp-mail.org/en/ — a temporary email address is generated automatically when the page loads. No sign-up required. Copy the email address shown on screen.

Step 2 — Fill in the Splunk registration form:

Paste the temp-mail address into the Email field. Fill in the remaining fields (First Name, Last Name, Company, Job Title, Phone) with any dummy information — for example: John Smith, Home Lab, Student, 555-0100. These fields are not verified.

Step 3 — Confirm your account:

Splunk will send a confirmation email. Go back to the temp-mail.org tab — the email will appear in the inbox automatically within a minute. Click the confirmation link inside it. You can now log in and access the download page.

Deploy on an Azure VM (Recommended)

Running Splunk on an Azure VM means it stays running when your laptop is closed, you can access it from anywhere, and you can send logs directly from your Active Directory VM (Lab 1) to the same Splunk instance over the internal Azure network.

VM Setting	Value
OS	Ubuntu 22.04 LTS (free tier eligible)
Size	Standard_B2s (2 vCPU, 4GB RAM minimum — Splunk requires at least 4GB)
Disk	30GB minimum
Inbound NSG ports	8000 (Splunk web UI) · 9997 (forwarder input) · 22 (SSH)

Open port 8000 and port 22 to your IP address only. Open port 9997 to the VNet address range only (10.0.0.0/16 or similar) so only your other Azure VMs can send logs to it — not the public internet.

How to SSH into your Linux VM

SSH (Secure Shell) is how you connect to and control a Linux VM remotely using a terminal. The tool you use to SSH depends on your operating system.

Mac or Linux — no extra tools needed

SSH is built into macOS and Linux. Open Terminal and run:

```
ssh yourusername@YOUR_VM_PUBLIC_IP
```

Replace yourusername with the admin username you set when creating the VM in Azure, and YOUR_VM_PUBLIC_IP with the public IP address shown in the Azure portal. Type yes when prompted about the host fingerprint, then enter your password.

Windows — install PuTTY first

Windows does not have a built-in SSH client that is easy to use for this purpose. PuTTY is the most widely used free SSH client for Windows. Download and install it before proceeding.

Installing PuTTY:

5. Go to putty.org and click Download PuTTY
6. Under Package files, download the 64-bit Windows installer (.msi file)
7. Run the installer and accept all defaults
8. Open PuTTY from the Start menu when installation is complete

Connecting with PuTTY:

9. Open PuTTY — the configuration window opens automatically

10. In the Host Name (or IP address) field, enter your VM's public IP address from the Azure portal
11. Make sure Port is set to 22 and Connection type is SSH (both are the defaults)
12. Click Open — a security alert about the server's host key will appear the first time. Click Accept.
13. A terminal window opens. Type your VM admin username and press Enter, then type your password and press Enter
14. You are now connected. The commands in the code block below run inside this PuTTY terminal window

Note: when you type your password in PuTTY, nothing appears on screen — no dots, no asterisks. This is normal Linux behaviour. Type your password and press Enter even though the cursor does not move.

```
# SSH into your Ubuntu VM, then run these commands:

# Download Splunk Enterprise — current version 10.2.2 as of April 2026
wget -O splunk-10.2.2-linux-amd64.deb
"https://download.splunk.com/products/splunk/releases/10.2.2/linux/splunk-10.2.2-80b90d638de6-linux-amd64.deb"
# NOTE: Splunk updates the URL with every new release. If this wget returns a 404 error,
# log into splunk.com, go to Free Trials and Downloads, select Linux .deb,
# and copy the wget command shown on the download page for the current version.

# Install the package
sudo dpkg -i splunk-10.2.2-linux-amd64.deb

# Start Splunk and accept the licence — you will be prompted to set admin credentials
sudo /opt/splunk/bin/splunk start --accept-license

# Enable Splunk to start automatically when the VM reboots
sudo /opt/splunk/bin/splunk enable boot-start

# Open the web UI in your browser:
# http://<YOUR_VM_PUBLIC_IP>:8000
```

Step 2 — Configure Data Inputs

Splunk is only as useful as the data it receives. The first data source to configure is your Active Directory / Windows Server VM from Lab 1. You will install the Splunk Universal Forwarder on that Windows Server and point it at your Splunk instance.

Part A — Enable receiving in Splunk first

Before installing the forwarder, tell Splunk to listen for incoming data on port 9997.

15. Log into the Splunk web UI at <http://your-vm-ip:8000>
16. Click Settings in the top navigation bar → Forwarding and Receiving
17. Click Configure Receiving → New Receiving Port → enter 9997 → Save
18. Click Settings → Indexes → Create New Index → name it windows_logs → Save

Part B — Install the Universal Forwarder on Windows Server

Do this on your Windows Server VM from Lab 1 — not on the Splunk Ubuntu VM.

19. On your Windows Server VM, open a browser and go to splunk.com/en_us/download/universal-forwarder.html
20. Download the Windows 64-bit installer
21. Run the installer — when asked for a Deployment Server, enter your Splunk VM's private IP address and port 8089
22. When asked for a Receiving Indexer, enter your Splunk VM's private IP address and port 9997
23. Complete the installation with the default settings

Part C — Configure inputs.conf to specify which logs to forward

The inputs.conf file tells the forwarder exactly which Windows Event Logs to collect and send. You need to create or edit this file on your Windows Server VM.

The file path below must be exact. Navigate to this folder in Windows Explorer: C:\Program Files\SplunkUniversalForwarder\etc\system\local\ — if the local folder does not exist, create it. Open Notepad as Administrator and save the file to that location.

```
# File location on Windows Server VM:
# C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf

# Each section in square brackets defines one log source to collect

[WinEventLog://Security]
# The Security log contains all authentication events – logins, failures, lockouts
disabled = 0
# disabled=0 means enabled. disabled=1 would turn this source off.
start_from = oldest
# Collect historical events, not just new ones going forward
current_only = 0
evt_resolve_ad_obj = 1
# Resolve Active Directory object names so usernames appear instead of SIDs

[WinEventLog://System]
# System log contains OS-level events – service starts/stops, driver failures
disabled = 0

[WinEventLog://Application]
# Application log contains events from installed applications
disabled = 0
```

```
# After saving inputs.conf, restart the forwarder to apply the changes
# Run this in PowerShell as Administrator on the Windows Server VM:
Restart-Service SplunkForwarder
```

If you do not have the Lab 1 AD VM yet: Splunk comes with sample data built in. Go to Search → Data Summary and look for sample indexes. You can complete most of the search exercises below using the built-in sample data while you build out Lab 1.

Step 3 — Essential SPL Searches

All searches are typed into the search bar at the top of the Search & Reporting app. Select a time range using the time picker on the right side of the search bar. Each search below is explained so you understand what every part does.

Confirm data is flowing

```
index=windows_logs | head 100
# index=windows_logs - search only in the windows_logs index you created
# | head 100          - return only the first 100 events
# If this returns results, your forwarder is working correctly
# If it returns nothing, check that the SplunkForwarder service is running on the
Windows VM
```

Find failed login attempts (EventCode 4625)

```
index=windows_logs sourcetype=WinEventLog:Security EventCode=4625
| stats count by Account_Name, Workstation_Name
| sort -count

# EventCode=4625          - filter to failed logon events only
# | stats count by ...    - count the events grouped by username and source
machine
# | sort -count           - sort highest count first (the minus sign means
descending)
# A count of 5+ for one account in a short window is a possible brute force
attempt
```

Find successful logins (EventCode 4624)

```
index=windows_logs sourcetype=WinEventLog:Security EventCode=4624
| stats count by Account_Name, Logon_Type
| sort -count

# Logon_Type 2 = interactive login (someone sat at the keyboard)
# Logon_Type 3 = network login (accessing a file share or network resource)
# Logon_Type 10 = remote interactive (RDP session)
# Logon_Type 5 = service account login (automated, usually expected)
```

Find account lockout events (EventCode 4740)

```
index=windows_logs sourcetype=WinEventLog:Security EventCode=4740
| table _time, Account_Name, Caller_Computer_Name
| sort -_time

# EventCode=4740          - account lockout events only
# | table _time, ...      - display as a table with these three columns
# _time                  - the timestamp of the event
# Caller_Computer_Name    - the machine where the failed attempts came from
# Multiple lockouts for the same account = likely brute force or password spray
```

Top 10 failed login usernames — threat hunting

```
index=windows_logs sourcetype=WinEventLog:Security EventCode=4625 earliest=-24h
| stats count as failures by Account_Name
| sort -failures
| head 10

# earliest=-24h          - only look at the last 24 hours
# count as failures      - rename the count column to 'failures' for readability
# | head 10              - return only the top 10 results
# Accounts with 20+ failures in 24 hours warrant investigation
# Usernames that do not exist in AD indicate an account enumeration attack
```

Detect after-hours logins

```
index=windows_logs sourcetype=WinEventLog:Security EventCode=4624
| eval hour=strftime(_time, "%H")
| where hour < 7 OR hour > 19
| table _time, Account_Name, Workstation_Name, Logon_Type
| sort -_time

# | eval hour=strftime(_time, "%H") - extract just the hour from the timestamp
# | where hour < 7 OR hour > 19      - keep only events outside 7am-7pm
# After-hours service account logins (Type 5) are normal and expected
# After-hours interactive logins (Type 2 or 10) from regular users warrant review
```

Step 4 — Build a Security Dashboard

Dashboards give you a permanent view of your security posture without running searches manually every time. Build one now using the searches from Step 3.

24. In Splunk, click Dashboards in the top navigation
25. Click Create New Dashboard
26. Name it: Windows Security Overview → click Create Dashboard
27. Click Add Panel for each panel below

28. For each panel, select New Search, paste the search, choose the visualisation type, and save

Panel	Search to use	Visualisation
Failed Logins — Last 24h	EventCode=4625 with stats count by Account_Name	Bar chart
Account Lockouts — Last 7d	EventCode=4740 with table output	Events list
Login Activity Over Time	EventCode=4624 with timechart count	Line chart
Top Source IPs — After Hours	After-hours search with stats count by Workstation_Name	Column chart

Step 5 — Create an Automated Alert

Alerts automate detection. Instead of checking dashboards manually, Splunk runs a search on a schedule and notifies you when the conditions are met. This is how real SOC detection works.

Run this search in the search bar first to confirm it works:

```
index=windows_logs sourcetype=WinEventLog:Security EventCode=4625
| stats count as failures by Account_Name
| where failures > 10
# This finds any account with more than 10 failed logins
# In a real environment, 10 failures in 15 minutes is a strong brute force
indicator
```

29. Click Save As → Alert
30. Name: Potential Brute Force — High Failure Count
31. Alert type: Scheduled
32. Run every: 15 minutes
33. Trigger condition: Number of Results is greater than 0
34. Trigger actions: Add to Triggered Alerts
35. Click Save

The quality of your alerts determines the quality of your security monitoring. An alert that fires too broadly creates alert fatigue — analysts stop paying attention. An alert that is too narrow misses real threats. Setting the threshold at 10 failures is a starting point. In a real environment you would tune this based on how many false positives you see over time.

Verification — Confirm the Lab is Working

Check	How to verify
Data is flowing into Splunk	Run: <code>index=windows_logs head 10</code> — should return recent events
Failed login search works	Run the <code>EventCode=4625</code> search — if no results, intentionally type the wrong password on your Windows VM a few times then search again
Dashboard displays data	Windows Security Overview dashboard shows populated panels with charts
Alert is active	Settings → Searches, Reports, and Alerts — your alert appears as Enabled

Portfolio entry for this lab. Take screenshots of your dashboard and your alert configuration. Export a few interesting searches as reports. Write a one-paragraph summary of what each panel shows and why it matters. This is concrete evidence of SIEM experience before your first job.